```
> restart;# The modified Planck's law
> restart;#Planck  for γ - ray
```

$$> \lambda 2 := 345.35\cdot 10^{-9};\ \lambda 1 := 7.8\cdot 10^{-10};\ c := 299792458;\ v1 := \frac{c}{\lambda 1};\ E1 := 3.59;$$

$$E3 := 2.446\cdot 10^{6};\ \lambda 3 := \frac{E1}{E3}\cdot 7.8\cdot 10^{-10};\ v3 := \frac{c}{\lambda 3};\ rb := 5.292410932\ 10^{-11};$$

$$\lambda 2 := 3.453500000 \times 10^{-7}$$

$$\lambda 1 := 7.800000000 \times 10^{-10}$$

$$c := 299792458$$

$$v1 := 3.843493051 \times 10^{17}$$

$$E1 := 3.59$$

$$E3 := 2.446000000 \times 10^{6}$$

$$\lambda 3 := 1.144807849 \times 10^{-15}$$

$$v3 := 2.618714208 \times 10^{23}$$

$$rb := 5.292410932 \times 10^{-11} \quad \textbf{(1)}$$

$$> E\gamma := \mathrm{diff}\left(Tq\cdot\frac{32\cdot\pi^{2}\cdot\nu^{2}}{2\cdot 3}\cdot E0,\ \mathrm{nu}\right);\ ;\ Tq := 1.765358264\times 10^{-19}$$

#The calculus $= 8\cdot Pi^{2}\cdot\frac{\nu^{2}}{c^{2}}\ \frac{4\cdot\pi\cdot rb^{3}}{3\cdot Tq}\cdot U(\nu, t)$ thus for γ

−ray the internal energy $U(\nu, t)$ is constant #'$E0$

' because the energy of a transversal wave is constant like the initial oscillation of one nucleus

and depend only of number of states $32\cdot Pi^{2}\cdot\frac{rb^{3}\cdot\nu^{2}}{3\cdot c^{2}\cdot Tq}$;

$$E\gamma := \frac{32\ Tq\ \pi^{2}\ \nu\ E0}{3}$$

$$Tq := 1.765358264 \times 10^{-19} \quad \textbf{(2)}$$

```
>
```

$$> E\gamma := \frac{32\ \ Tq\ \pi^{2}\ \nu\ \ E0}{3};$$

$$E\gamma := 1.858494688 \times 10^{-17}\ \nu\ E0 \quad \textbf{(3)}$$

$$> c := 299792458;\ Sq := 4.25\cdot 10^{-19};\ \ h := 4.135667697\times 10^{-15};\ R := 3.566639512\times 10^{22};\ Tq := 1.765358264\times 10^{-19};\ ec := 1.602176634$$

$\times 10^{-19}$; $v3 := \frac{c}{\lambda 3}$; $v1 := \frac{c}{\lambda 1}$; #$Vg := \frac{R}{2\ Tq}$;

$$c := 299792458$$

$$Sq := 4.250000000 \times 10^{-19}$$

$$h := 4.135667697 \times 10^{-15}$$

$$R := 3.566639512 \times 10^{22}$$

$$Tq := 1.765358264 \times 10^{-19}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$v3 := 2.618714208 \times 10^{23}$$

$$v1 := 3.843493051 \times 10^{17} \quad \textbf{(4)}$$

> $E\gamma := \frac{32\ Tq\ \pi^2\ \nu\ E0}{3}$;

$$E\gamma := 1.858494688 \times 10^{-17}\ \nu\ E0 \quad \textbf{(5)}$$

> $\lambda min := evalf\left(\frac{c}{10^{23}}\right)$;

Thus the minimum wavelenght is ~ times the nucleon radius as in plot below

$$\lambda min := 2.997924580 \times 10^{-15} \quad \textbf{(6)}$$

> $h\gamma := \frac{E1}{v1}$;

$$h\gamma := 9.340461794 \times 10^{-18} \quad \textbf{(7)}$$

> $solve(3.59 = g \cdot v1)$; $h\gamma := 9.340461795 \times 10^{-18}$; $subs(\nu = 3.843493051 \times 10^{17}, h\gamma \cdot \nu)$; $solve\left(h\gamma = \frac{32\ Tq\ \pi^2\ E0}{3}, E0\right)$

$$9.340461794 \times 10^{-18}$$

$$h\gamma := 9.340461795 \times 10^{-18}$$

$$3.590000000$$

$$0.5025821088 \quad \textbf{(8)}$$

> $E1 := \frac{1239.84}{\lambda 2 \cdot 10^9}$;

The Energy of 3.59eV for photons of wave-lenght 345.35 nm;

$$E1 := 3.590097003 \quad \textbf{(9)}$$

```
> E3 := 2.446·10^6 ; λ3 := (E1/E3)·7.8·10^-10; v3 := c/λ3 ; λ1 := 7.8·10^-10; v1 := c/λ1 ;
```

$$E3 := 2.446000000 \times 10^{6}$$

$$\lambda 3 := 1.144838783 \times 10^{-15}$$

$$v3 := 2.618643450 \times 10^{23}$$

$$\lambda 1 := 7.800000000 \times 10^{-10}$$

$$v1 := 3.843493051 \times 10^{17} \quad \textbf{(10)}$$

```
> E:=he*nu;
```

$$E := he\,\nu \quad \textbf{(11)}$$

```
> #The Planck 's law for γ−ray : hγ =9.340461795×10^-18 in eV·s ;
> hγ := 9.3404617950×10^-18; ec := 1.602176634·10^-19; c := 299792458;
  h := 6.62607015·10^-34/ec ; he := 4.13566733·10^-15; λ1 := 7.8·10^-10 ; v1 := c/7.8·10^-10# h 4.135667697×10^-15eV·s & hγ = 9.3404617950×10^-18eV·s;
```

$$h\gamma := 9.340461795 \times 10^{-18}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$c := 299792458$$

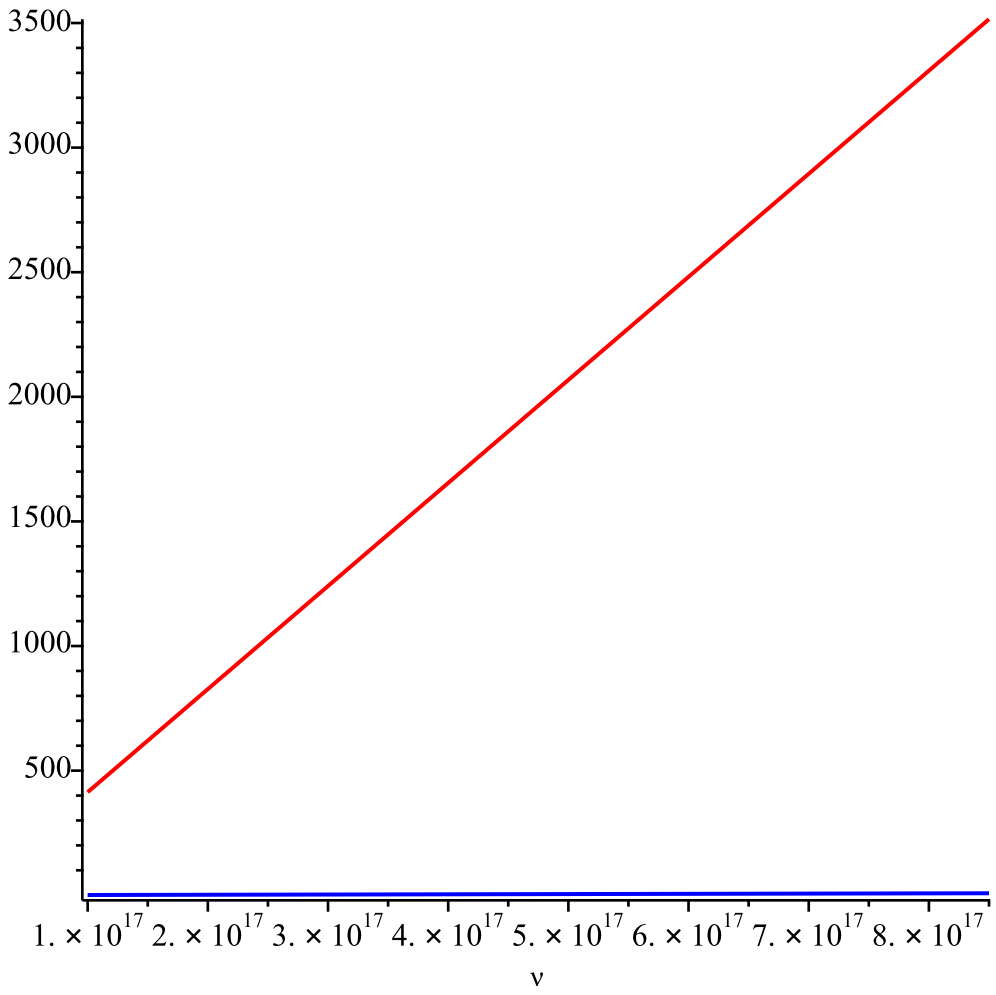
$$h := 4.135667697 \times 10^{-15}$$

$$he := 4.135667330 \times 10^{-15}$$

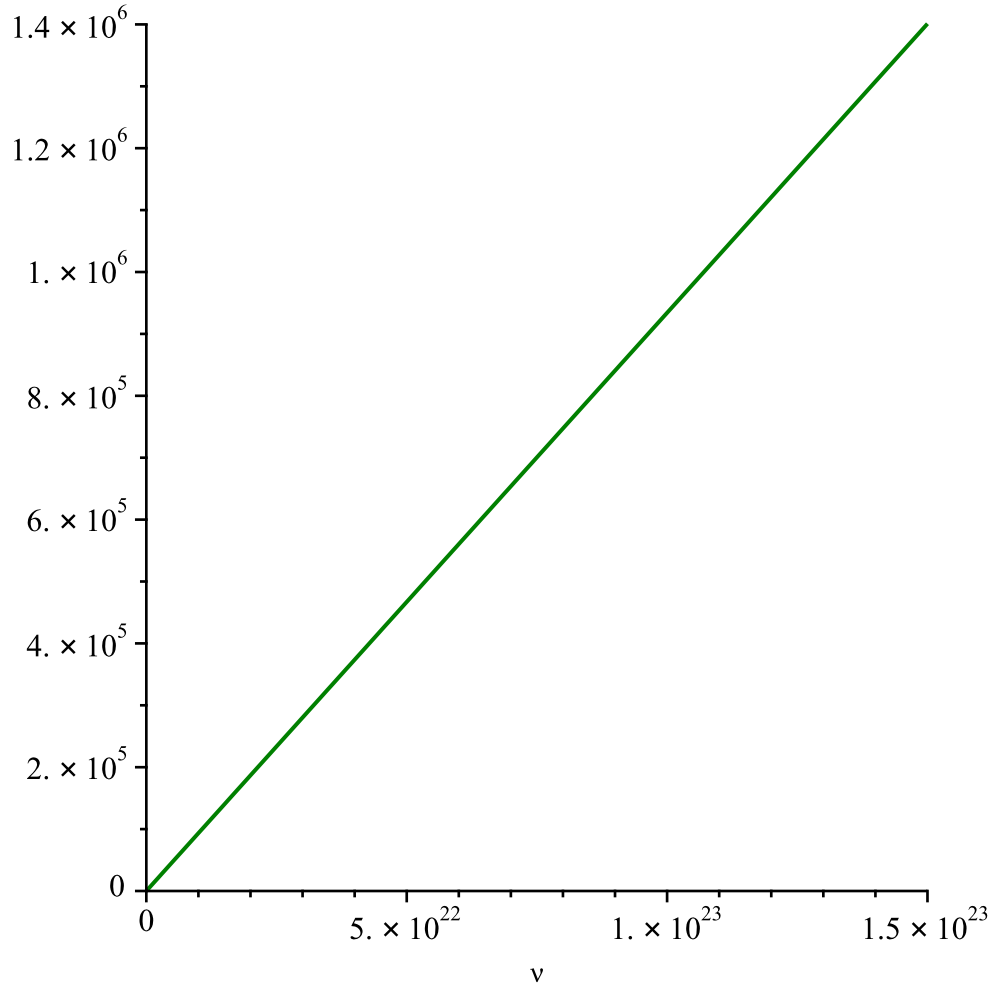
$$\lambda 1 := 7.800000000 \times 10^{-10}$$

$$v1 := 3.843493051 \times 10^{17} \quad \textbf{(12)}$$

```
> plot([9.3404617950×10^-18·ν, E], ν= 10^17 .. 8.5·10^17, color=["Blue", "Red"]);
  #The Planck's law for γ - ray liniar approx. "Blue" the real energy and Red with photon Planck constant which does not match the real energies!!
```




```
> plot(9.3404617950×10^-18·ν, ν= 3·10^17 ... 15·10^22, color = "Green" );
    #The Planck's law for γ - ray "Green" with` `γ — ray Planck
    liniar the real value of energy
```




```
> #  Thus, is a huge difference in the calculus of the gamma-ray
  energy with the two equations.
> restart;#The best fit for photons Planck's low
> E(nu):=Em+E0*ln(1+2.787176283*10^(-17)*nu)+C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) + C \nu^2 \quad (13)$$

```
> E1:=subs(nu=9.733521364*10^16,E(nu));
```

$$E1 := Em + E0 \ln\left(3.712903990\right) + 9.474143814 \times 10^{33} C \quad (14)$$

```
> E2:=subs(nu=7.880979442*10^14,E(nu));
```

$$E2 := Em + E0 \ln\left(1.021965679\right) + 6.210983697 \times 10^{29} C \quad (15)$$

```
> E3:=subs(nu=8.689636465*10^14,E(nu));
```

$$E3 := Em + E0 \ln\left(1.024219549\right) + 7.550978189 \times 10^{29} C \quad (16)$$

```
> solve({E1=3.259,E2=3.259,E3=3.59});
```

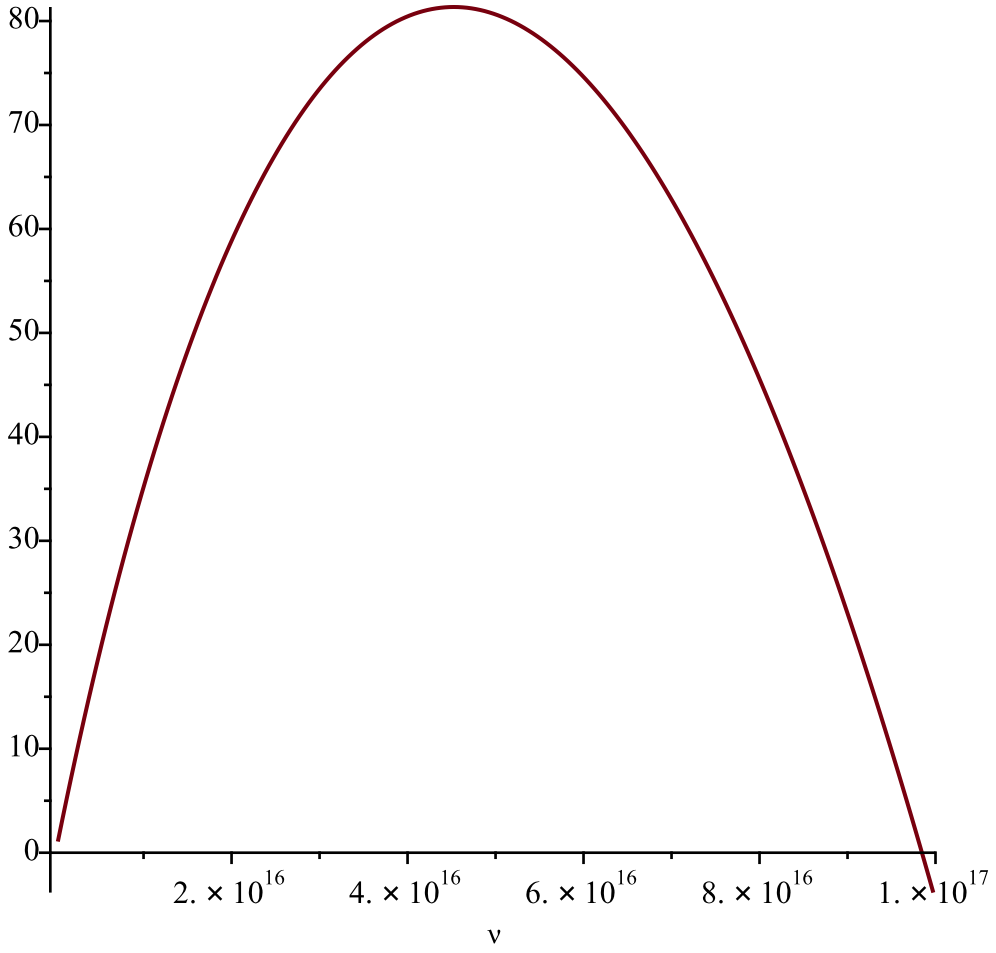
$$\left\{C = -2.063162888 \times 10^{-32}, E0 = 151.5047392, Em = -0.02006692051\right\} \quad (17)$$

```
> E(nu):=subs(Em=-0.02006692051,E0=151.5047392,C=-2.063162888*10^
  (-32),E(nu));
```

(18)

$$E(\nu) := -0.02006692051 + 151.5047392 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) - 2.063162888 \times 10^{-32} \nu^2 \quad \textbf{(18)}$$

```
> plot(E(nu),nu=10^13..10^17, caption = typeset("Graph of the
  Planck's constant [eV] as a function of λ in [nm] "));
```



Graph of the Planck's constant [eV] as a function of λ in [nm]

```
> he:=6.6260701500*10^(-34)/1.6021766340/10^(-19);
```

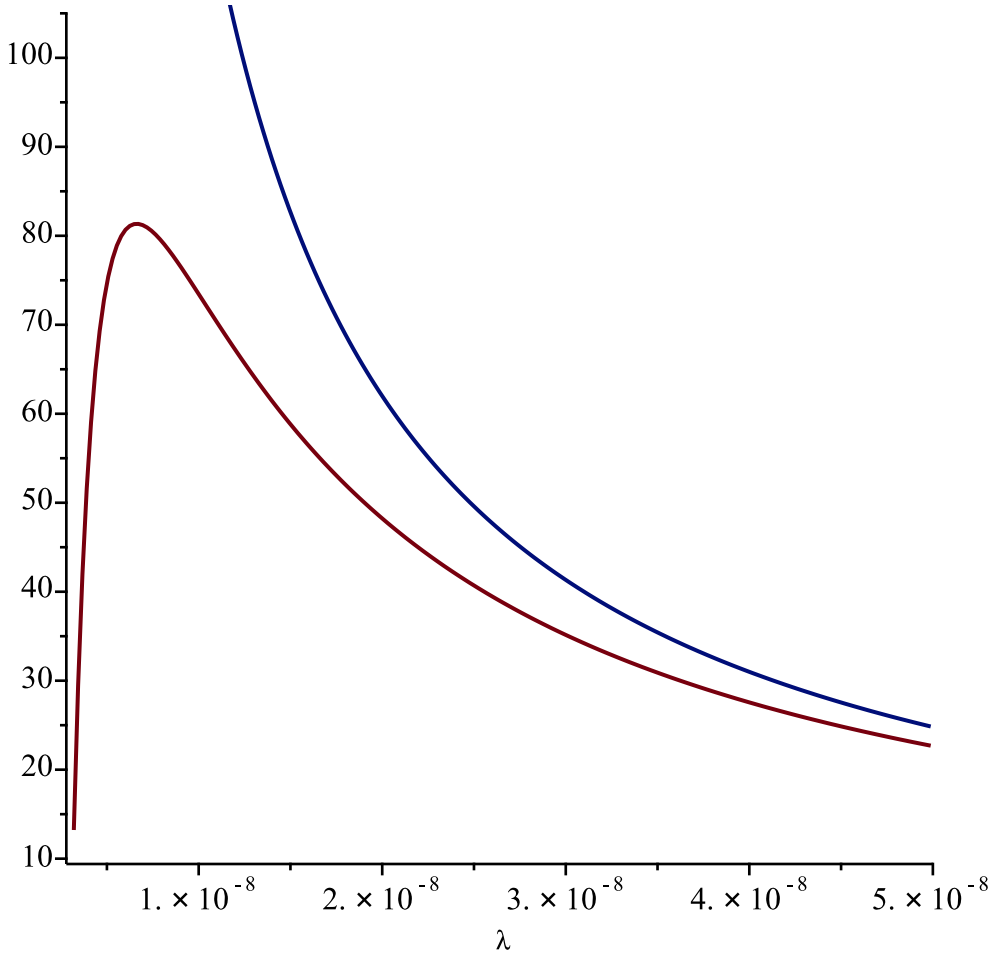
$$he := 4.135667697 \times 10^{-15} \quad \textbf{(19)}$$

```
> c := 299792458;B := lambda -> subs(nu = c/lambda, -0.02006692051
  + 151.5047392*ln(1 + 2.787176283*1/100000000000000000*nu) + (-1)*
  2.063162888*1/10000000000000000000000000000000*nu^2);plot([B
  (lambda),he*c/lambda], lambda = 3.08*10^(-9) .. 5*10^(-8),caption
  = typeset("A plot of Energy theoretical and practical in [eV]
  versus lambda in [nm]"));
```

$$c := 299792458$$

$$B := \lambda \mapsto subs\left(\nu = \frac{c}{\lambda}, -0.02006692051 + 151.5047392 \cdot \ln\left(1 + 2.787176283 \times 10^{-17} \cdot \nu\right)\right.$$

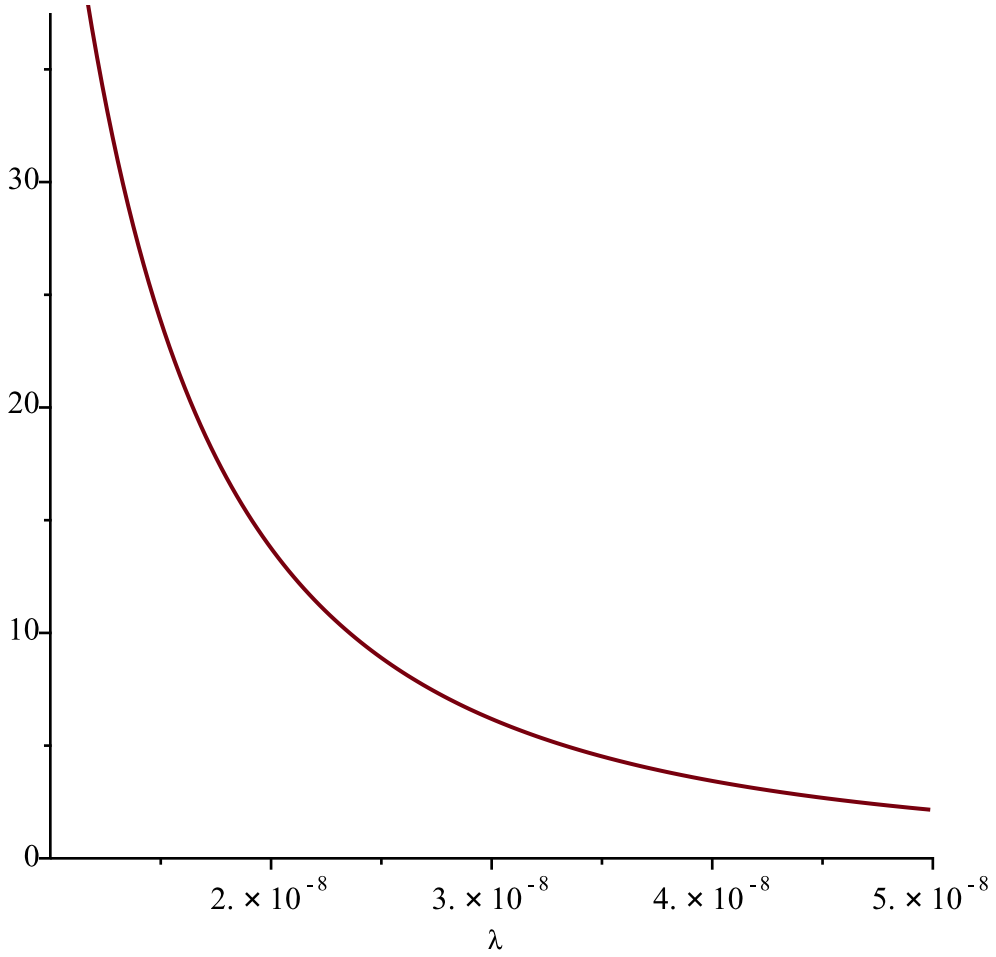
$$-2.063162888 \times 10^{-32} \cdot \nu^2\Bigg)$$



A plot of Energy theoretical and practical in [eV] versus lambda in [nm]

```
> -B(lambda)+he*c/lambda;plot(%,lambda = 3.08*10^(-9) .. 5*10^(-8),
  caption = typeset("Graph of the error [eV] as a function of λ in
  [nm] between the two calculation methods "));# Below the
  wavelength of 30 nm, the error becomes larger and below 15-20nm
  the error is dominant.
```

$$0.02006692051 - 151.5047392 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) + \frac{1.854278330 \times 10^{-15}}{\lambda^2} + \frac{1.239841984 \times 10^{-6}}{\lambda}$$



Graph of the error [eV] as a function of λ in [nm] between the two calculation

```
> solve(diff(B(lambda), lambda) = 0);# @ lambda= 6.625nm the
  photons have maximum of energy
```

$$-3.695060778 \times 10^{-9},\ 6.624561165 \times 10^{-9} \qquad \textbf{(20)}$$

```
> h := 6.6260701500*10^(-34);ec := 1.6021766340*10^(-19);rb:=5.239*
  10^(-13);he:=h/ec;
  c := 299792458;T := 273 + 24;k := 1.3806490000*10^(-23);Tq:=
  1.765*10^(-19);nu0:=evalf(c/(3.804*10^(-7)));nu1:=evalf(c/(3.08*
  10^(-9)));nu2:= evalf(c/(3.135*10^(-7)))
```

$$h := 6.626070150 \times 10^{-34}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$rb := 5.239000000 \times 10^{-13}$$

$$he := 4.135667697 \times 10^{-15}$$

$$c := 299792458$$

$$T := 297$$

$$k := 1.380649000 \times 10^{-23}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$v0 := 7.880979442 \times 10^{14}$$

$$v1 := 9.733521364 \times 10^{16}$$

$$v2 := 9.562757830 \times 10^{14} \quad (21)$$

```
> evalf(subs(nu=7.880979442*10^14,E(nu)));evalf(subs(nu=c/(380.43*
  10^(-9)),E(nu)));
```

$$3.258999999$$

$$3.258745253 \quad (22)$$

```
> evalf(subs(nu=8.689636465*10^14,E(nu)));evalf(subs(nu=c/(345.35*
  10^(-9)),E(nu)));
```

$$3.590000000$$

$$3.586400619 \quad (23)$$

```
> evalf(subs(nu=9.733521364*10^16,E(nu)));nu1:=evalf(c/(3.08*10^
  (-9)));
```

$$3.2590000$$

$$v1 := 9.733521364 \times 10^{16} \quad (24)$$

```
> evalf(subs(nu=nu2,E(nu)));evalf(subs(nu=c/(17*10^(-9)),E(nu)));
```

$$3.946259923$$

$$54.13412164 \quad (25)$$

```
> solve(-0.02006692051 + 151.5047392*ln(1 + 2.787176283*10^(-17)*
  nu) - 2.063162888*10^(-32)*nu^2=5.28,nu)
```

Warning, solutions may have been lost

```
> evalf(subs(nu=5*10^16,E(nu)));# ~The Energy peak in eV
```

$$80.63322078 \quad (26)$$

```
> evalf(subs(nu=0.984563*10^17,E(nu)));c/(0.984563*10^17);
```

$$0.0000560$$

$$3.044929150 \times 10^{-9} \quad (27)$$

```
> 1239.84/8.54;nu3:=c/(%*10^(-9));
```

$$145.1803279$$

$$v3 := 2.064966117 \times 10^{15} \quad (28)$$

```
> evalf(subs(nu=nu3,E(nu)));8.54-%;# @ 145nm the error is 0.16eV;
```

$$8.370000712$$

$$0.169999288 \quad (29)$$

```
> 1239.84/5.32;nu3:=c/(%*10^(-9));
```

$$233.0526316$$

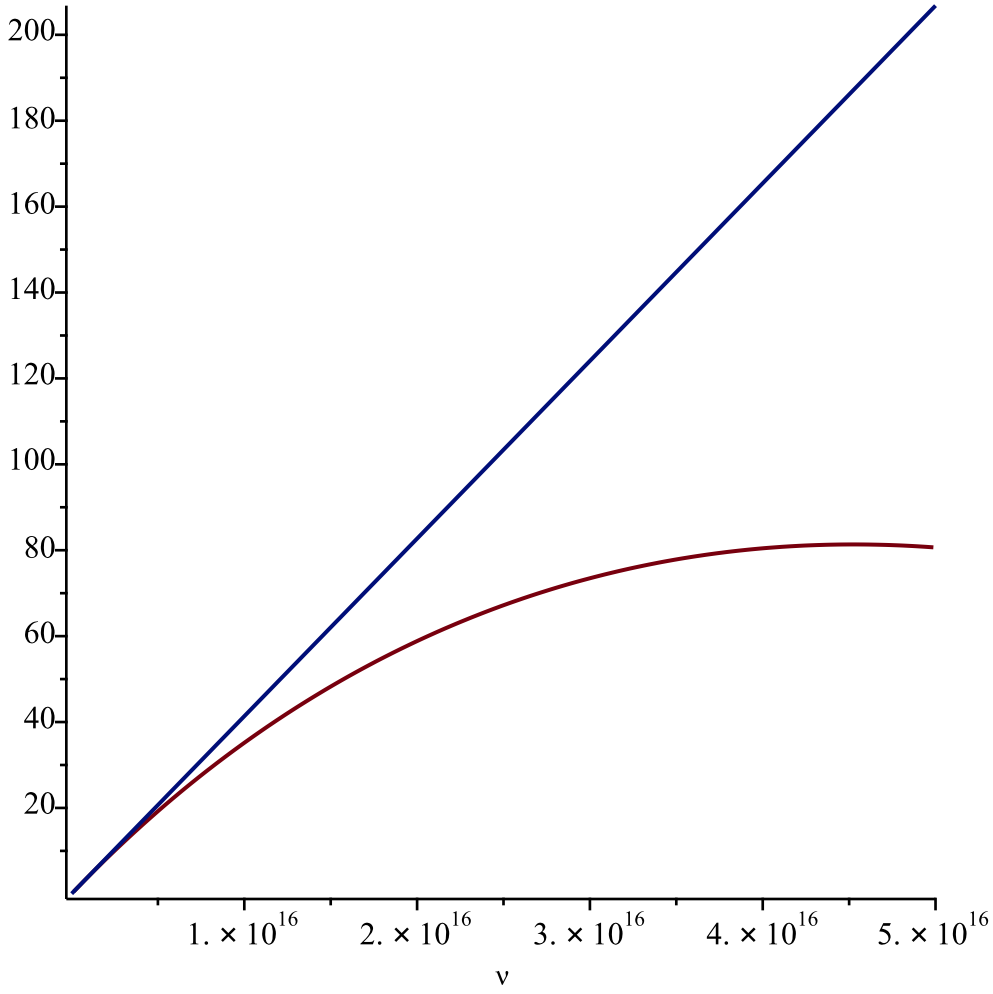
$$v3 := 1.286372335 \times 10^{15} \quad (30)$$

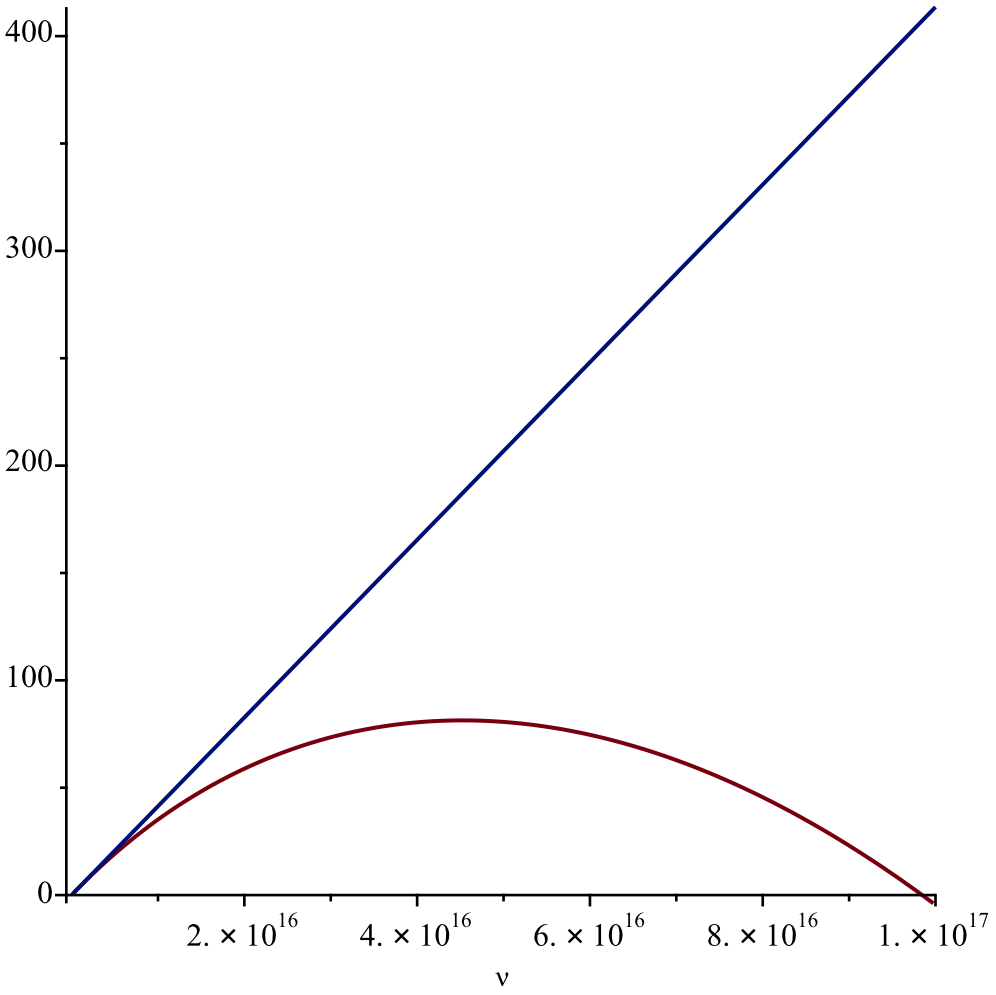
```
> evalf(subs(nu=nu3,E(nu)));5.32-%;#@ 233nm the error is 0.037eV;
```

$$5.282651909$$

$$0.037348091 \quad (31)$$

```
> plot([E(nu),he*nu], nu=10^13..5*10^16);plot([E(nu),he*nu], nu=
  10^13..10^17);
```

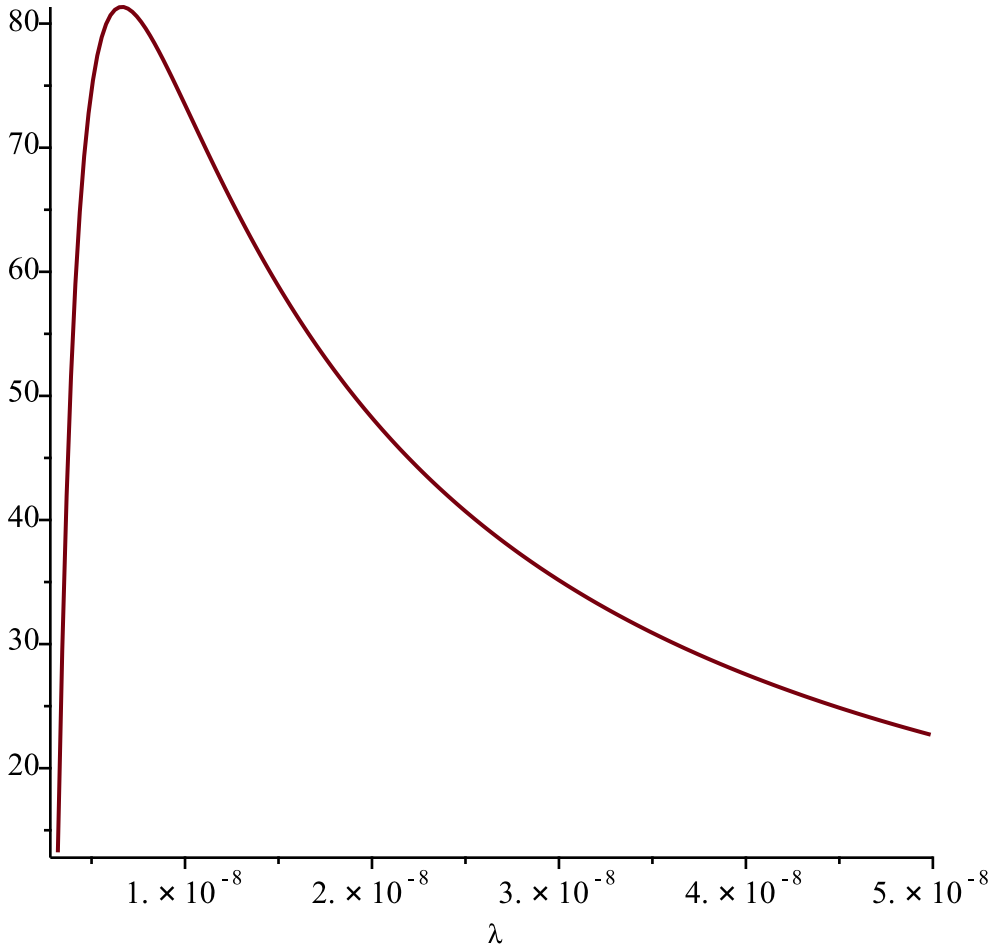





```
> c := 299792458;B := lambda -> subs(nu = c/lambda, -0.02006692051
  + 151.5047392*ln(1 + 2.787176283*1/10000000000000000*nu) + (-1)*
  2.063162888*1/100000000000000000000000000000000*nu^2);plot(B
  (lambda), lambda = 3.08*10^(-9) .. 5*10^(-8),caption = typeset("A
  plot of Energy [eV] versus lambda in [nm]"));
```

$$c := 299792458$$

$$B := \lambda \mapsto subs\left(\nu = \frac{c}{\lambda},\ -0.02006692051 + 151.5047392 \cdot \ln\left(1 + 2.787176283 \times 10^{-17} \cdot \nu\right) - 2.063162888 \times 10^{-32} \cdot \nu^{2}\right)$$



A plot of Energy [eV] versus lambda in [nm]

```
> B(lambda);
```

$$-0.02006692051 + 151.5047392 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) - \frac{1.854278330 \times 10^{-15}}{\lambda^2} \quad (32)$$

```
> solve(diff(B(lambda), lambda) = 0);# @ lambda= 6.625nm the
  photons have maximum of energy
```

$$-3.695060778 \times 10^{-9},\ 6.624561165 \times 10^{-9} \quad (33)$$

```
>
> restart;#The Phonon Planck's "constant"
> Digits:=20;
```

$$Digits := 20 \quad (34)$$

```
> E(nu):=Em+E0*ln(1+2*Tq1*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, A\, Tq1\, \nu + 1) + C\, \nu^2 \quad (35)$$

```
> c:=299792458;nu1:=435;ec:= 1.602176634*10^(-19);E1:=3.259;T:=297;
  va:=343;Tq1:=c/va*1.765*10^(-19);ha:=0.007486790719; ha*ec;Tq1*
  105.2757803;2*Tq1*32/3*Pi^2;
```

$$c := 299792458$$

$$vl := 435$$

$$ec := 1.6021766340000000000 \times 10^{-19}$$

$$E1 := 3.259$$

$$T := 297$$

$$va := 343$$

$$Tq1 := 1.5426638144897959184 \times 10^{-13}$$

$$ha := 0.007486790719$$

$$1.1995161153629859846 \times 10^{-21}$$

$$1.6240513681098771170 \times 10^{-11}$$

$$3.2481027355498213704 \times 10^{-11} \quad (36)$$

```
> E(nu):=Em+E0*ln(1+A*3.248102735549821370 4*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(1 + 3.2481027355498213704 \times 10^{-11} A \nu\right) + C \nu^2 \quad (37)$$

```
> E1:=subs(nu=435,E(nu));
```

$$E1 := Em + E0 \ln\left(1 + 1.4129246899641722961 \times 10^{-8} A\right) + 189225\, C \quad (38)$$

```
> E2:=subs(nu=528,E(nu));
```

$$E2 := Em + E0 \ln\left(1 + 1.7149982443703056836 \times 10^{-8} A\right) + 278784\, C \quad (39)$$

```
> E3:=subs(nu=25000,E(nu));
```

$$E3 := Em + E0 \ln\left(1 + 8.1202568388745534260 \times 10^{-7} A\right) + 625000000\, C \quad (40)$$

```
> solve({E1=3.259,E2=3.95});
```

$$\{A = A, C = 0.00001116582364698131957 7\, E0 \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right) \quad (41)$$
$$- 0.00001116582364698131957 7\, E0 \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right)$$
$$+ 7.7155841400640918277 \times 10^{-6}, E0 = E0, Em = 2.1128529796000401970\, E0 \ln\left(1.$$
$$+ 1.7149982443703056836 \times 10^{-8} A\right) - 3.1128529796000401970\, E0 \ln\left(1.$$
$$+ 1.4129246899641722961 \times 10^{-8} A\right) + 1.7990185910963722239\}$$

```
> sol:=solve({E1=3.259,E2=3.95,E3=0},{Em,E0,C,A});
```

$$sol := \left\{A = A, C = -\left(0.0010000000000000000000\right.\right. \left(-3259. \ln\left(1. + 1.7149982443703056836\right.\right. \quad (42)$$
$$\left.\times 10^{-8} A\right) - 691. \ln\left(1. + 8.1202568388745534260 \times 10^{-7} A\right) + 3950. \ln\left(1.$$
$$\left.\left.\left.+ 1.4129246899641722961 \times 10^{-8} A\right)\right)\right) \Big/ \left(89559. \ln\left(1. + 8.1202568388745534260\right.\right.$$
$$\left.\left.\times 10^{-7} A\right) - 6.24810775 \times 10^{8} \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right) + 6.24721216\right.$$

$$\times 10^{8} \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right)\Big), E0 = -\left(4.3203611830600000000\right.$$

$$\times 10^{8}\Big) \Big/ \left(89559. \ln\left(1. + 8.1202568388745534260 \times 10^{-7} A\right) - 6.24810775 \times 10^{8} \ln\left(1.\right.\right.$$

$$+ 1.7149982443703056836 \times 10^{-8} A\Big) + 6.24721216 \times 10^{8} \ln\left(1.\right.$$

$$+ 1.4129246899641722961 \times 10^{-8} A\Big)\Big), Em$$

$$= \left(0.0020000000000000000000 \left(8.0559153 \times 10^{7} \ln\left(1. + 8.1202568388745534260 \times 10^{-7} A\right)\right.\right.$$

$$- 1.018437500000 \times 10^{12} \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right) + 1.234375000000$$

$$\times 10^{12} \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right)\Big)\Big) \Big/ \left(89559. \ln\left(1.\right.\right.$$

$$+ 8.1202568388745534260 \times 10^{-7} A\Big) - 6.24810775 \times 10^{8} \ln\left(1.\right.$$

$$+ 1.7149982443703056836 \times 10^{-8} A\Big) + 6.24721216 \times 10^{8} \ln\left(1.\right.$$

$$\left.\left.\left.+ 1.4129246899641722961 \times 10^{-8} A\Big)\Big)\right\}\right.$$

```
> A:=9.949747*10^6;evalf(sol[4]);#Em
```

$$A := 9.949747000000 \times 10^{6}$$

$$Em = -0.26049922890091899376 \quad (43)$$

```
> Em := -0.26049922890091899376;Em;
```

$$Em := -0.26049922890091899376$$

$$-0.26049922890091899376 \quad (44)$$

```
> evalf(sol[3]);#E0
```

$$E0 = 26.892256305433408378 \quad (45)$$

```
> E0 := 26.892256305433408378;E0;
```

$$E0 := 26.892256305433408378$$

$$26.892256305433408378 \quad (46)$$

```
> evalf(sol[2]);;#C
```

$$C = -9.4502695574974682733 \times 10^{-8} \quad (47)$$

```
> C:=-9.4502695574974682733*10^(-8);
```

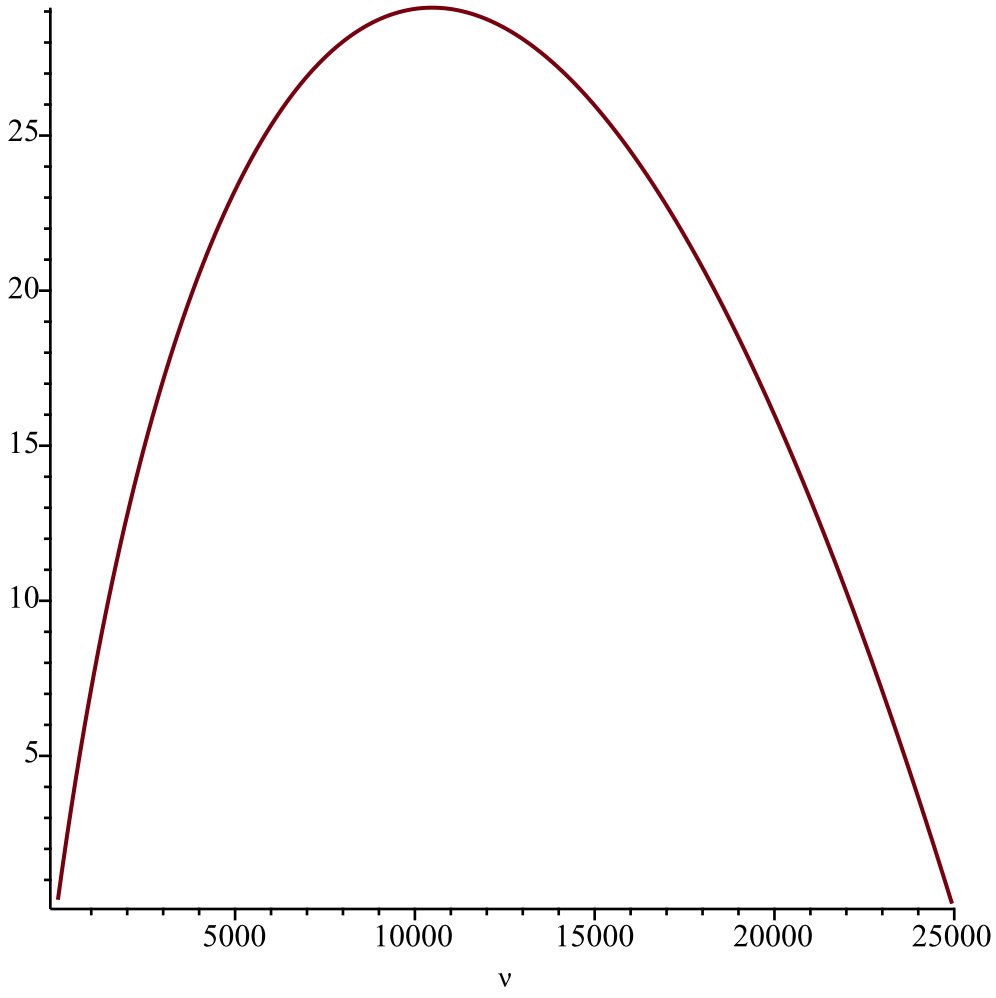
$$C := -9.4502695574974682733 \times 10^{-8} \quad (48)$$

```
> E(nu):=Em+E0*ln(1+A*3.2481027355498213704*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := -0.26049922890091899376 + 26.892256305433408378 \ln\left(1\right.$$

$$\left.+ 0.00032317800448728628531 \nu\right) - 9.4502695574974682733 \times 10^{-8} \nu^{2} \quad (49)$$

```
> plot(E(nu),nu=10..25000);
```




```
> evalf(subs(nu=435,E(nu)));evalf(subs(nu=528,E(nu)));evalf(subs
  (nu=480,E(nu)));
```

3.2589999999999993

3.9499999999999993

3.5958312874552960791 **(50)**

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30,E(nu)));evalf(subs(nu=
  13350,E(nu)));#The max value is ~26.7eV at frequency ~13350Hz
  [30Hz÷24khz]
```

−0.26049922890091899376

−0.0011105298534884395 5

27.818493430720448758 **(51)**

```
> restart;#The momopole RadioFrequency Planck's "constant"
> Digits:=25;
```

$$Digits := 25 \quad \textbf{(52)}$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, A\, Tq\, \nu + 1) + C\, \nu^2 \tag{53}$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);
  ec:= 1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \tag{54}$$

```
> A0:=64/3*Tq;
```

$$A0 := 3.765333333333333333333333 \times 10^{-18} \tag{55}$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 3.765333333333333333300000 \times 10^{-18}\, A\, \nu) + C\, \nu^2 \tag{56}$$

```
> E1:=subs(nu=5.754173859*10^8,E(nu));
```

$$E1 := Em + E0 \ln(1 + 2.166638263708799999980819 \times 10^{-9}\, A) + 3.311051679959895188100000 \times 10^{17}\, C \tag{57}$$

```
> E2:=subs(nu=6.39691958*10^8,E(nu));
```

$$E2 := Em + E0 \ln(1 + 2.408653452522666666645344 \times 10^{-9}\, A) + 4.092058011298737640000000 \times 10^{17}\, C \tag{58}$$

```
> E3:=subs(nu=31*10^9,E(nu));
```

$$E3 := Em + E0 \ln(1 + 1.167253333333333333323000 \times 10^{-7}\, A) + 96100000000000000000\, C \tag{59}$$

```
> solve({E1=3.259,E2=3.59});
```

$$\{A = A,\ C = 1.280399351290465204177160 \times 10^{-17}\, E0 \ln(1. + 2.166638263708799999980819 \times 10^{-9}\, A) - 1.280399351290465204177160 \times 10^{-17}\, E0 \ln(1. + 2.408653452522666666645344 \times 10^{-9}\, A) + 4.238121852771439825826398 \times 10^{-18},\ E0 = E0,\ Em = 4.239468423109854807131941\, E0 \ln(1. + 2.408653452522666666645344 \times 10^{-9}\, A) - 5.239468423109854807131941\, E0 \ln(1. + 2.166638263708799999980819 \times 10^{-9}\, A) + 1.855735951950638058839327\} \tag{60}$$

```
> sol:=solve({E1=3.259,E2=3.59,E3=0},{Em,E0,C,A});
```

$$sol := \{A = A,\ C = -(0.10000000000000000000000000\ (-3259. \ln(1. + 2.408653452522666666645344 \times 10^{-9}\, A) - 331. \ln(1. + 1.167253333333333333323000 \tag{61}$$

$$\times 10^{-7} A) + 3590. \ln(1. + 2.16663826370879999998081 9 \times 10^{-9} A)))/$$

$$(7.810063313388424519 \times 10^{18} \ln(1. + 1.16725333333333333323000 \times 10^{-7} A)$$

$$- 9.6066889483200401048119 \times 10^{22} \ln(1. + 2.40865345252266666666645344 \times 10^{-9} A)$$

$$+ 9.6059079419887012623600 \times 10^{22} \ln(1. + 2.16663826370879999998081 9 \times 10^{-9} A)),$$

$$E0 = -(3.1823593415277665622434 81 \times 10^{22}) / (7.810063313388424519 \times 10^{18} \ln(1.$$

$$+ 1.16725333333333333323000 \times 10^{-7} A) - 9.6066889483200401048119 \times 10^{22} \ln(1.$$

$$+ 2.40865345252266666666645344 \times 10^{-9} A) + 9.6059079419887012623600 \times 10^{22} \ln(1.$$

$$+ 2.16663826370879999998081 9 \times 10^{-9} A)), Em$$

$$= (0.01000000000000000000000000 (1.449341527766562243481 \times 10^{21} \ln(1.$$
$$+ 1.16725333333333333323000 \times 10^{-7} A) - 3.131899000000000000000000 \times 10^{25} \ln(1.$$
$$+ 2.40865345252266666666645344 \times 10^{-9} A) + 3.449990000000000000000000 \times 10^{25} \ln(1.$$
$$+ 2.16663826370879999998081 9 \times 10^{-9} A)))/(7.810063313388424519 \times 10^{18} \ln(1.$$
$$+ 1.16725333333333333323000 \times 10^{-7} A) - 9.6066889483200401048119 \times 10^{22} \ln(1.$$
$$+ 2.40865345252266666666645344 \times 10^{-9} A) + 9.6059079419887012623600 \times 10^{22} \ln(1.$$
$$+ 2.16663826370879999998081 9 \times 10^{-9} A))\}$$

```
> A:=6993.8*10^4;evalf(sol[4]);#Em
```

$$A := 6.99380000 \times 10^{7}$$

$$Em = 0.036003711081316326770796 41 \quad (62)$$

```
> Em:=0.036003711081316326770796 41;
```

$$Em := 0.036003711081316326770796 41 \quad (63)$$

```
> evalf(sol[3]);#E0=1.722290354;
```

$$E0 = 22.967590176857168364806 45 \quad (64)$$

```
> E0:=22.967590176857168364806 45;
```

$$E0 := 22.967590176857168364806 45 \quad (65)$$

```
> evalf(sol[2]);#C
```

$$C = -5.298079951889684414200720 \times 10^{-20} \tag{66}$$

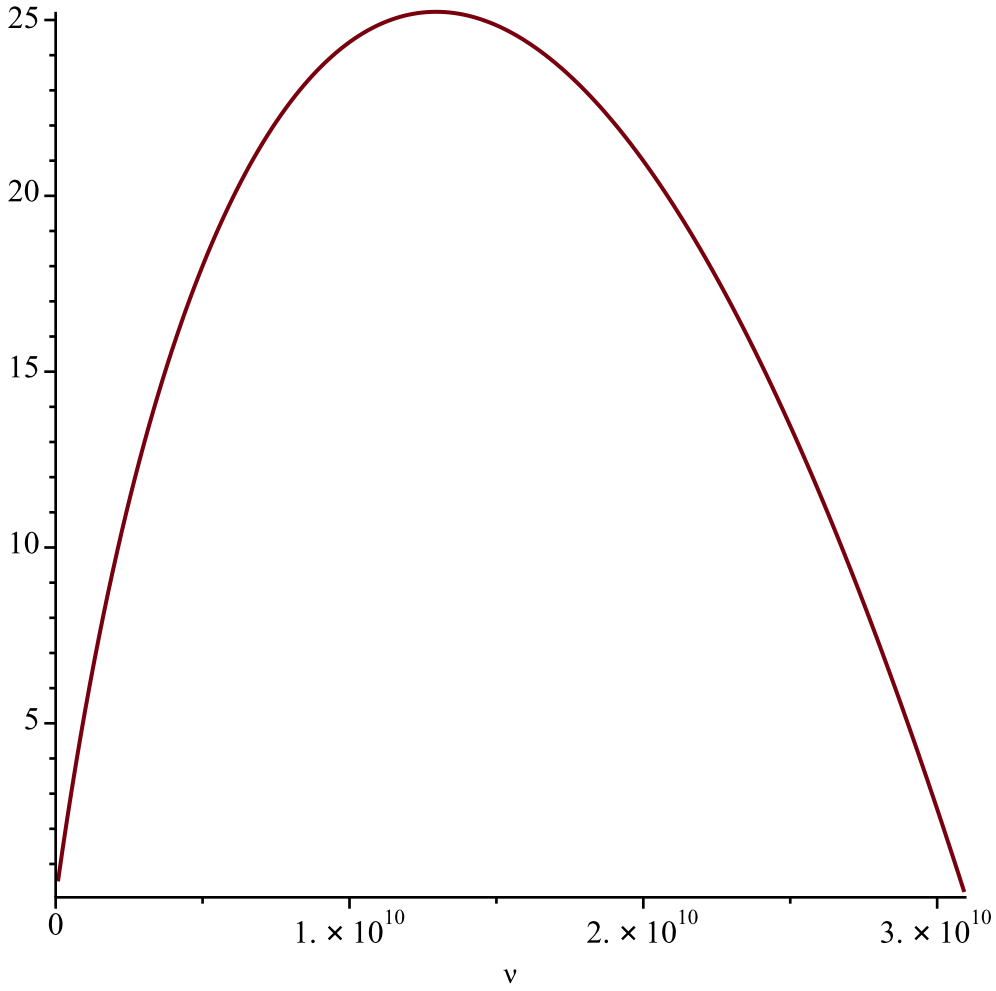
```
> C:=-5.29807995188968441420072*10^(-20);
```

$$C := -5.298079951889684414200720 \times 10^{-20} \tag{67}$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := 0.03600371108131632677079641 + 22.967590176857168364806 45 \ln(1 + 2.633398826666666666643354 \times 10^{-10} \nu) - 5.298079951889684414200720 \times 10^{-20} \nu^2 \tag{68}$$

```
> plot(E(nu),nu=10^6..3.1*10^10);
```




```
> evalf(subs(nu=5.754173859*10^8,E(nu)));evalf(subs(nu=6.39691958*
  10^8,E(nu)));evalf(subs(nu=7.111105195*10^8,E(nu)));
```

$$3.259000000000000000000005$$

$$3.590000000000000000000005$$

$$3.951629162260248123877006 \tag{69}$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=1*10^3,E(nu)));evalf(subs
  (nu=14.91*10^9,E(nu)));#24.9eV @ 14.91Ghz in the range
  [1Khz-31Ghz]
```

$$0.03600371108131632677079641$$

$$0.03600975936296927876148310$$

$$24.882246359040279205451 50 \quad (70)$$

```
> mu:=4*Pi*10^(-7);rb:= 5.292410932*10^(-11);solve((Pi*mu*E0m*rb^3)
  ^0.5=2.360457908*10^(-18));
```

$$\mu := \frac{\pi}{2500000}$$

$$rb := 5.29241093200000000000000 \times 10^{-11}$$

$$9.52077080897457803652756 6 \quad (71)$$

```
> h := 6.62607015*10^(-34);ec:= 1.602176634*10^(-19);he:=h/ec;rm:=
  5.64*10^(-21);Bmw:=mu*B*h/(8*Pi*rm^2);mu*B*he/(8*Pi*rm^2);E0m:=
  24.689224432880536497102 86;Bmw/1.748;(8*Pi/3*mu*E0m*rb^3)^0.5/
  (Pi*mu*E0m*rb^3)^0.5;
```

$$h := 6.62607015000000000000000 \times 10^{-34}$$

$$ec := 1.60217663400000000000000 \times 10^{-19}$$

$$he := 4.13566769692385864616223 1 \times 10^{-15}$$

$$rm := 5.64000000000000000000000 \times 10^{-21}$$

$$Bmw := 3.84633523589576731552738 9 \times 10^{-18}$$

$$24.006936278261609671729 80$$

$$E0m := 24.689224432880536497102 86$$

$$2.20042061550101105007287 7 \times 10^{-18}$$

$$1.63299316185545206546485 6 \quad (72)$$

```
> (Pi*mu*E0m*rb^3)^0.5;(8*Pi/3*mu*E0m*1.478*rb^3)^0.5;(8*Pi/3*mu*
  E0m*rb^3)^0.5/(Pi*mu*E0m*rb^3)^0.5;Bm:=Bmw/1.478;Bm/1.478;
```

$$3.80114055570773170171654 5 \times 10^{-18}$$

$$7.54632512009842862860317 6 \times 10^{-18}$$

$$1.63299316185545206546485 6$$

$$Bm := 2.60239190520687910387509 4 \times 10^{-18}$$

$$1.76075230392887625431332 5 \times 10^{-18} \quad (73)$$

```
>
```

```
> restart;#The Microwave Planck's low
```

```
> Digits:=25;
```

$$Digits := 25 \quad (74)$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, A\, Tq\, \nu + 1) + C\, \nu^2 \tag{75}$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);
  ec:= 1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \tag{76}$$

```
> E(nu):=Em+E0*ln(1+A*2.82400000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{77}$$

```
> A0:=16*Tq;
```

$$A0 := 2.824000000000000000000000 \times 10^{-18} \tag{78}$$

```
> E(nu):=Em+E0*ln(1+A*2.82400000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{79}$$

```
> E1:=subs(nu=160.23*10^9,E(nu));
```

$$E1 := Em + E0 \ln(1 + 4.524895200000000000000000 \times 10^{-7} A) + 2.567365290000000000000000 \times 10^{22} C \tag{80}$$

```
> E2:=subs(nu=285*10^9,E(nu));
```

$$E2 := Em + E0 \ln(1 + 8.048400000000000000000000 \times 10^{-7} A) + 8122500000000000000000 C \tag{81}$$

```
> E3:=subs(nu=6.85*10^12,E(nu));
```

$$E3 := Em + E0 \ln(1 + 0.000019344400000000000000000000 A) + 4.692250000000000000000000 \times 10^{25} C \tag{82}$$

```
> solve({E1=7.15,E2=12.64});
```

$$\{A = A,\ C = 1.800136364289894960981063 \times 10^{-23} E0 \ln(1. + 4.524895200000000000000000 \times 10^{-7} A) - 1.800136364289894960981063 \times 10^{-23} E0 \ln(1. + 8.048400000000000000000000 \times 10^{-7} A) + 9.882748639951523335786037 \times 10^{-23},\ E0 = E0,\ Em = 0.462160761894467182056869 E0 \ln(1. + 8.048400000000000000000000 \times 10^{-7} A) - 1.462160761894467182056869 E0 \ln(1. + 4.524895200000000000000000 \times 10^{-7} A) + 4.612737417199375170507791\} \tag{83}$$

```
> sol:=solve({E1=7.15,E2=12.64,E3=0},{Em,E0,C,A});
```

$$sol := \left\{A = A,\ C = \left(1.000000000000000000000000 \times 10^{-16} \left(715. \ln(1.\right.\right.\right. \tag{84}$$

$$+ 8.048400000000000000000000 \times 10^{-7} A) + 549. \ln(1.$$

$$+ 0.000019344400000000000000000000 A) - 1264. \ln(1. + 4.52489520000000000000000000$$

$$\times 10^{-7} A)))/(5.55513471 \times 10^{8} \ln(1. + 0.000019344400000000000000000000 A)$$

$$- 4.68968263471 \times 10^{11} \ln(1. + 8.048400000000000000000000 \times 10^{-7} A)$$

$$+ 4.68412750000 \times 10^{11} \ln(1. + 4.52489520000000000000000000 \times 10^{-7} A)), E0 =$$

$$-(2.57860768777344000000000 \times 10^{12})/(5.55513471 \times 10^{8} \ln(1.$$

$$+ 0.000019344400000000000000000000 A) - 4.68968263471 \times 10^{11} \ln(1.$$

$$+ 8.048400000000000000000000 \times 10^{-7} A) + 4.68412750000 \times 10^{11} \ln(1.$$

$$+ 4.52489520000000000000000000 \times 10^{-7} A)), Em$$

$$= (0.640000000000000000000000 (4.003809021 \times 10^{9} \ln(1.$$
$$+ 0.000019344400000000000000000000 A) - 5.242123046875 \times 10^{12} \ln(1.$$
$$+ 8.048400000000000000000000 \times 10^{-7} A) + 9.267193750000 \times 10^{12} \ln(1.$$
$$+ 4.52489520000000000000000000 \times 10^{-7} A)))/(5.55513471 \times 10^{8} \ln(1.$$
$$+ 0.000019344400000000000000000000 A) - 4.68968263471 \times 10^{11} \ln(1.$$
$$+ 8.048400000000000000000000 \times 10^{-7} A) + 4.68412750000 \times 10^{11} \ln(1.$$
$$+ 4.52489520000000000000000000 \times 10^{-7} A))\}$$

```
> A:=79904.7*10^1;evalf(sol[4]);#Em
```

$$A := 799047.0$$

$$Em = -1.978297828703956720793870 \quad \textbf{(85)}$$

```
> Em:=-1.978297828703956720793870;
```

$$Em := -1.978297828703956720793870 \quad \textbf{(86)}$$

```
> evalf(sol[3]);#E0=28.87280028
```

(87)

$$E0 = 29.720763361662431074128 13 \quad (87)$$

```
> E0:=29.72076336166243107412813;
```

$$E0 := 29.720763361662431074128 13 \quad (88)$$

```
> evalf(sol[2]);#C
```

$$C = -1.731841026798320557990073 \times 10^{-24} \quad (89)$$

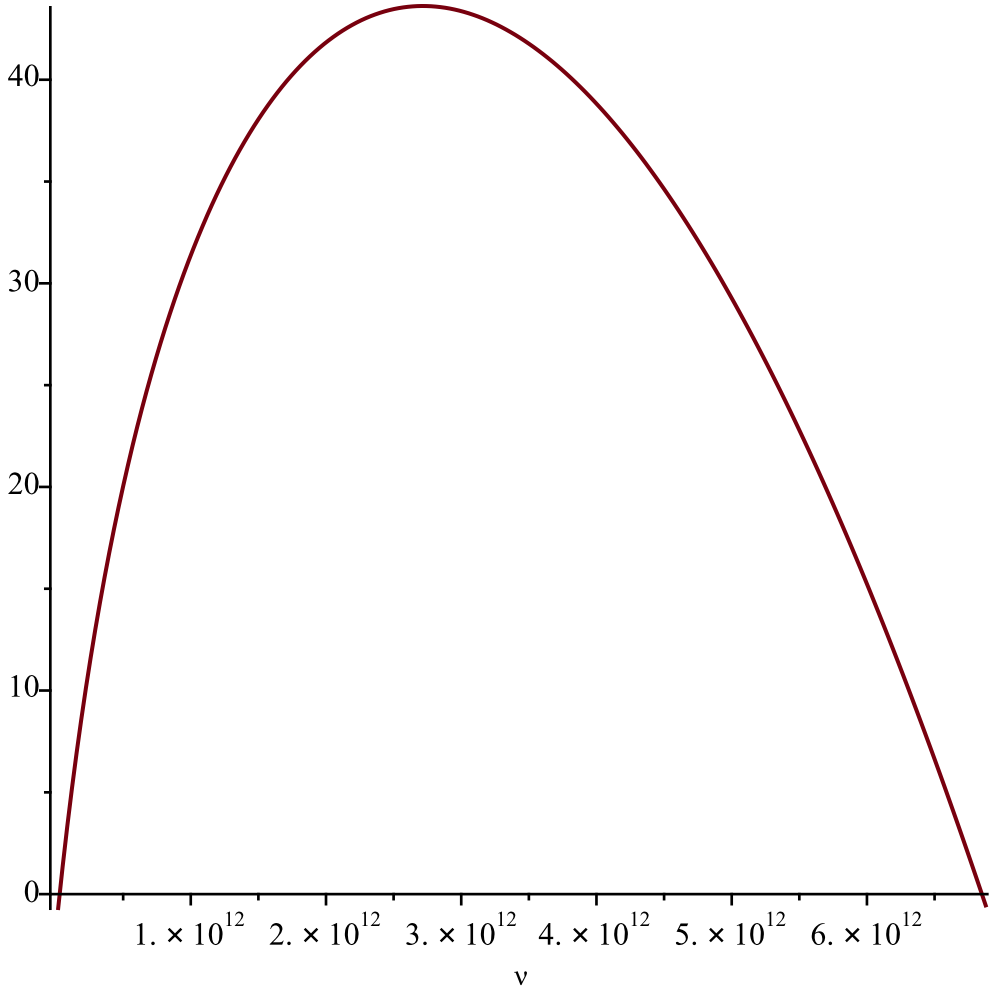
```
> C:=-1.731841026798320557990073*10^(-24);
```

$$C := -1.731841026798320557990073 \times 10^{-24} \quad (90)$$

```
> E(nu):=Em+E0*ln(1+A*2.824*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := -1.978297828703956720793870 + 29.720763361662431074128 13 \ln(1 + 2.256508728000000000000000 \times 10^{-12} \nu) - 1.731841026798320557990073 \times 10^{-24} \nu^2 \quad (91)$$

```
> plot(E(nu),nu=10^9..6.9*10^12);
```




```
> evalf(subs(nu=285*10^9,E(nu)));evalf(subs(nu=160.23*10^9,E(nu)));
  evalf(subs(nu=5.85*10^12,E(nu)));
```

12.640000000000000000000001

7.150000000000000000000002

(92)

$$17.611354876254039343043 45 \quad (92)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30.4*10^9,E(nu)));evalf
  (subs(nu=2.79*10^12,E(nu)));#43.9eV @ 15.4Ghz [30Ghz÷6.85Thz]
```

$$-1.9782978287039567207938 70$$

$$-0.0080034598257155563489 14$$

$$43.60434402635505383696412 \quad (93)$$

```
> c := 299792458;c/(6.85*10^12)
```

$$c := 299792458$$

$$0.0000437653223357664233576 6422 \quad (94)$$

```
> solve(12=160.23*10^9*hmw);#The energy of CMB
```

$$7.4892342258004119078824 19 \times 10^{-11} \quad (95)$$

```
> hmw:=7.4892342258004119078824 19*10^(-11);
```

$$hmw := 7.4892342258004119078824 19 \times 10^{-11} \quad (96)$$

```
> 160.23*10^9·hmw ; 285·10^9·hmw;
    #Penzias and Wilson originally detected the Cosmic Microwave
    Background 160.23÷285 GHz with an energy of ~12.64÷22.49 eV ;
```

$$12.000000000000000000000 0$$

$$21.344317543531173937464 89 \quad (97)$$

```
> # http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/h21.html
> #10.9Ghz is 3.59eV on the microwave thus we can deduce the
  Planck's constatnt for miccrowave;
> E:=hmw*nu;
```

$$E := 7.4892342258004119078824 19 \times 10^{-11} \nu \quad (98)$$

```
> hm1w=3.59/(10.9*10^9);#Posible value
```

$$hm1w = 3.2935779816513761467889 91 \times 10^{-10} \quad (99)$$

```
> 160.23*10^9 ;#Penzias and Wilson originally detected the Cosmic
  Microwave Background 160.23GHz with an energy of ~12eV ;
```

$$1.6023000000000 \times 10^{11} \quad (100)$$

```
> c := 299792458;
  160.23*10^9;
  evalf(c/(1.9*10^11));evalf(c/(10.9*10^9));
```

$$c := 299792458$$

$$1.6023000000000 \times 10^{11}$$

$$0.0015778550421052631578947 37$$

$$0.0275038952293577981651376 1 \quad (101)$$

```
> hm1w=0.188977/(1.057*10^9);13.60569292/4;3.5904-3.401423;#Posible
  value
```

$$hm1w = 1.78786187322611163670766 3 \times 10^{-10}$$

$$3.4014232300000000000000000$$

$$0.188977 \qquad \textbf{(102)}$$